

Question number	Answer	Additional guidance	Marks																																																																																																																																																										
	(radius calculated) $37 / 25$ (1) (radius converted into mm) $0.037 / 0.025$ (1) (mm³) expressed to max of 3 sig figures (1)	ECF if 62 is used ACCEPT standard form ECF for wrong conversion Correct bald answer = 3 marks, but with wrong order of magnitude = 2 marks or too many sig figs = 2 marks, or both order of magnitude and too many sig figs = 1 mark Correct bald answer for d = 62 = 2 marks, but with wrong order of magnitude = 1 mark or too many sig figs = 1 mark <table><tr><th>D</th><th>R (μm)</th><th>R(mm)</th><th>R3</th><th>π</th><th>4×π×R3</th><th>÷3</th><th>1 sig fig</th><th>2 sig figs</th><th>3 sig figs</th><th>Stand for</th></tr><tr><td></td><td>mp 1</td><td>mp 2</td><td></td><td></td><td></td><td></td><td>mp 3</td><td>mp 3</td><td>mp 3</td><td></td></tr><tr><td>74</td><td>37</td><td>0.037</td><td>0.000050653</td><td>3</td><td>0.000608</td><td>0.0002026</td><td>0.0002</td><td>0.00020</td><td>0.000203</td><td>10 -4</td></tr><tr><td>74</td><td>37</td><td>0.037</td><td>0.000050653</td><td>3.14</td><td>0.000636</td><td>0.000212067</td><td>0.0002</td><td>0.00021</td><td>0.000212</td><td>10 -4</td></tr><tr><td>74</td><td>37</td><td>0.037</td><td>0.000050653</td><td>3.142</td><td>0.000637</td><td>0.000212202</td><td>0.0002</td><td>0.00021</td><td>0.000212</td><td>10 -4</td></tr><tr><td>74</td><td>37</td><td>0.037</td><td>0.000050653</td><td>3.141593</td><td>0.000637</td><td>0.000212175</td><td>0.0002</td><td>0.00021</td><td>0.000212</td><td>10 -4</td></tr><tr><td>50</td><td>25</td><td>0.025</td><td>0.000015625</td><td>3</td><td>0.000188</td><td>0.0000625</td><td>0.00006</td><td>0.000063</td><td>0.0000625</td><td>10 -5</td></tr><tr><td>50</td><td>25</td><td>0.025</td><td>0.000015625</td><td>3.14</td><td>0.000196</td><td>6.54167E-05</td><td>0.00007</td><td>0.000065</td><td>0.0000654</td><td>10 -5</td></tr><tr><td>50</td><td>25</td><td>0.025</td><td>0.000015625</td><td>3.142</td><td>0.000196</td><td>6.54583E-05</td><td>0.00007</td><td>0.000065</td><td>0.0000655</td><td>10 -5</td></tr><tr><td>50</td><td>25</td><td>0.025</td><td>0.000015625</td><td>3.141593</td><td>0.000196</td><td>6.54498E-05</td><td>0.00007</td><td>0.000065</td><td>0.0000654</td><td>10 -5</td></tr><tr><td>62</td><td></td><td>0.031</td><td>0.000029791</td><td>3</td><td>0.000357</td><td>0.000119164</td><td>0.0001</td><td>0.00012</td><td>0.000119</td><td>10 -4</td></tr><tr><td>62</td><td></td><td>0.031</td><td>0.000029791</td><td>3.14</td><td>0.000374</td><td>0.000124725</td><td>0.0001</td><td>0.00012</td><td>0.000125</td><td>10 -4</td></tr><tr><td>62</td><td></td><td>0.031</td><td>0.000029791</td><td>3.142</td><td>0.000374</td><td>0.000124804</td><td>0.0001</td><td>0.00012</td><td>0.000125</td><td>10 -4</td></tr><tr><td>62</td><td></td><td>0.031</td><td>0.000029791</td><td>3.141593</td><td>0.000374</td><td>0.000124788</td><td>0.0001</td><td>0.00012</td><td>0.000125</td><td>10 -4</td></tr></table>	D	R (μm)	R(mm)	R3	π	4×π×R3	÷3	1 sig fig	2 sig figs	3 sig figs	Stand for		mp 1	mp 2					mp 3	mp 3	mp 3		74	37	0.037	0.000050653	3	0.000608	0.0002026	0.0002	0.00020	0.000203	10 -4	74	37	0.037	0.000050653	3.14	0.000636	0.000212067	0.0002	0.00021	0.000212	10 -4	74	37	0.037	0.000050653	3.142	0.000637	0.000212202	0.0002	0.00021	0.000212	10 -4	74	37	0.037	0.000050653	3.141593	0.000637	0.000212175	0.0002	0.00021	0.000212	10 -4	50	25	0.025	0.000015625	3	0.000188	0.0000625	0.00006	0.000063	0.0000625	10 -5	50	25	0.025	0.000015625	3.14	0.000196	6.54167E-05	0.00007	0.000065	0.0000654	10 -5	50	25	0.025	0.000015625	3.142	0.000196	6.54583E-05	0.00007	0.000065	0.0000655	10 -5	50	25	0.025	0.000015625	3.141593	0.000196	6.54498E-05	0.00007	0.000065	0.0000654	10 -5	62		0.031	0.000029791	3	0.000357	0.000119164	0.0001	0.00012	0.000119	10 -4	62		0.031	0.000029791	3.14	0.000374	0.000124725	0.0001	0.00012	0.000125	10 -4	62		0.031	0.000029791	3.142	0.000374	0.000124804	0.0001	0.00012	0.000125	10 -4	62		0.031	0.000029791	3.141593	0.000374	0.000124788	0.0001	0.00012	0.000125	10 -4	
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(3)

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Question number	Answer	Additional guidance	Mark
(i)	15 / 15.0	DO NOT ACCEPT any other values	(1)

Question number	Answer	Additional guidance	Mark
8(b)(ii)	A description that includes the following points: - credit for appropriate {sample to be used / details of method} (1) {stain / dye} {plates / (plates and) cells} (1) - credit what is being compared that would work (1)	e.g. cells and shed plates / two groups of cells / one large group of cells / put together in a tank of sea water / use a microscope ACCEPT marker / fluorescent tag IGNORE genetic profiling / gel electrophoresis / carbon dating NB if using unstained plates and unstained (naked) <i>E. huxley</i> this mark would be awarded for use of microscope and something else would be credited in mp 1 e.g. coloured plates attached to cells / cells different colour to plates ECF DNA profiling / gel electrophoresis : different bands on	

		year plates produced	
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Question number	Answer	Mark
(iii)	Effect of healthy <i>E. huxleyi</i>: • carbon dioxide absorbed from the oceans to make plates • carbon dioxide absorbed for photosynthesis Effect of marine snow : • {decomposers / increase in animals feeding on snow} respire • {releasing carbon dioxide / increasing carbon dioxide} levels in {water / air} • so more carbon dioxide will be released from water into air • plates sunk to the bottom of the ocean will act like a carbon sink • therefore carbon dioxide levels {will not increase / fall} in {water / air} • as {no / less} carbon dioxide will dissolve in the water from the air • {plates / <i>E. huxleyi</i>} stuck together will have a lower surface area • so will absorb less carbon dioxide from {water / air} • so less carbon dioxide will dissolve in the water from the air • marine snow will block light reaching {seaweeds / plants} • so less photosynthesis so less carbon dioxide removed from {water / air} • less carbon dioxide will be absorbed by the water from the air Effects on global warming / climate change: • more / less carbon dioxide accumulates in atmosphere so more / less infrared radiation will be trapped in the earth's atmosphere Implications of climate change : • flooding / melting of ice caps / enzyme denaturation	(6)

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	Simple descriptions 1 mark = one relevant comment 2 marks = description of global warming / effects on climate change / what healthy <i>E. huxleyi</i> do
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	Explanation of how {infected cells / marine snow} affects global warming / levels of carbon dioxide in air / water 3 marks = explanation of how {infected cells / marine snow} results in uptake or release of carbon dioxide 4 marks = plus a second reason for a change or the effect of a change on global warming or effect of a change on carbon dioxide levels {in water / in air}
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	Clear explanation of how the infected cells / marine snow affects climate change and its effects 5 marks = explanation includes either the implications of climate change or an understanding of how the levels of carbon dioxide in the water affect the carbon dioxide in the air 6 marks = explanation includes either the implications of climate change and an understanding of how the levels of carbon dioxide in the water affect the carbon dioxide in the air

